databricks

Lakeflow Jobs Core Components

LECTURE

Task Orchestration

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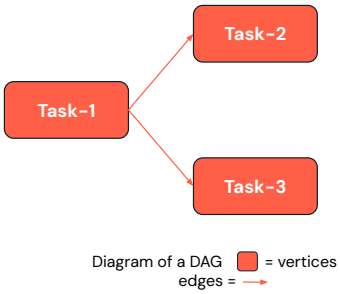
Now let's explore how tasks are orchestrated using DAG concepts.

Task Orchestration

What is a DAG?

A DAG is a **conceptual representation** of series of activities, including data processing flows.

- **D**irected - unambiguous direction for each edge
- **A**cyclic - contains no cycles
- **G**raph- collection of vertices connected by edges.



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A DAG is a conceptual representation of a series of activities, including data processing flows. Let's break down the acronym:

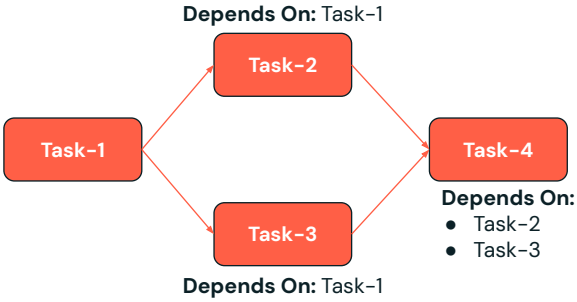
- Directed means there's an unambiguous direction for each edge - tasks flow in a specific direction.
- Acyclic means it contains no cycles - you can't have circular dependencies where Task A depends on Task B which depends on Task A.
- Graph means it's a collection of vertices connected by edges - in our case, vertices represent tasks and edges represent dependencies.

The diagram shows a simple DAG where Task-1 leads to both Task-2 and Task-3, representing the flow of execution.

Task Orchestration

Task Orchestration Overview

- Databricks Jobs supports **task orchestration**
  - The ability to run multiple tasks as a Directed Acyclic Graph (DAG)
  - Orchestrate tasks using the **Databricks UI** , **API, SDK** or **Databricks Asset Bundles**
- You can define the **order of execution of tasks** in a job by configuring task dependencies, creating a **DAG of task execution**



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Databricks Jobs supports task orchestration through the ability to run multiple tasks as a Directed Acyclic Graph (DAG). You can orchestrate tasks using the Databricks UI, API, SDK, or Databricks Asset Bundles.

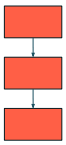
The example shows that Task-2 depends on Task-1, Task-3 also depends on Task-1, and Task-4 depends on both Task-2 and Task-3. You define the order of execution by configuring these task dependencies, creating a DAG of task execution.

This approach allows you to build complex workflows while maintaining clear execution order and dependencies.

Task Orchestration

Common Workloads Patterns

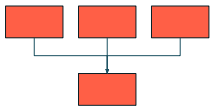
Sequence



Sequence

- Data transformation/ processing/cleaning
- Bronze/silver/gold tables

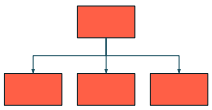
Funnel



Funnel

- Multiple data sources
- Data collection

Fan-out



Fan-out, star pattern

- Single data source
- Data ingestion and distribution

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There are three common workflow patterns you'll encounter:

- Sequence Pattern is used for data transformation, processing, cleaning, and building bronze/silver/gold tables in a medallion architecture.
- Funnel Pattern brings together multiple data sources for data collection and consolidation.
- Fan-out or Star Pattern takes a single data source and distributes it for data ingestion and distribution to multiple downstream systems.

Understanding these patterns helps you design effective workflows for your specific use cases.

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Thank you for completing this lesson and continuing your journey to develop your skills with us.